Jet completion: finite control and proposed next search

Research follow-up for PR 2969

2026-11-01

Evidence scope

The existing native exact-rational control at $L = 4, 8, 16, 32$ checks the original cardinal-polynomial kernel against its ordinary first-jet flattening control. Its recorded first-coefficient valuations differ by two at each size, while both normalized residue ranks stay 2. The four rows contain 5,440 nodal value/first-derivative checks in total. These are finite exact observations, not a runtime comparison or a uniform theorem.

Recorded finite observations

L	Nodal checks	Original v_2	Flattened v_2	Both residue ranks
4	64	3	5	2
8	256	4	6	2
16	1,024	5	7	2
32	4,096	6	8	2

The values are copied from `native/expected.txt` and match `native/replay.log`. The prospective protocol identifies the coefficients: the original is linear in the scaled chart and the flattened control is quadratic. Flattening roughly doubles polynomial degree and adds two valuation bits in these cases; the data do not establish that this trade is beneficial after degree and precision are priced.

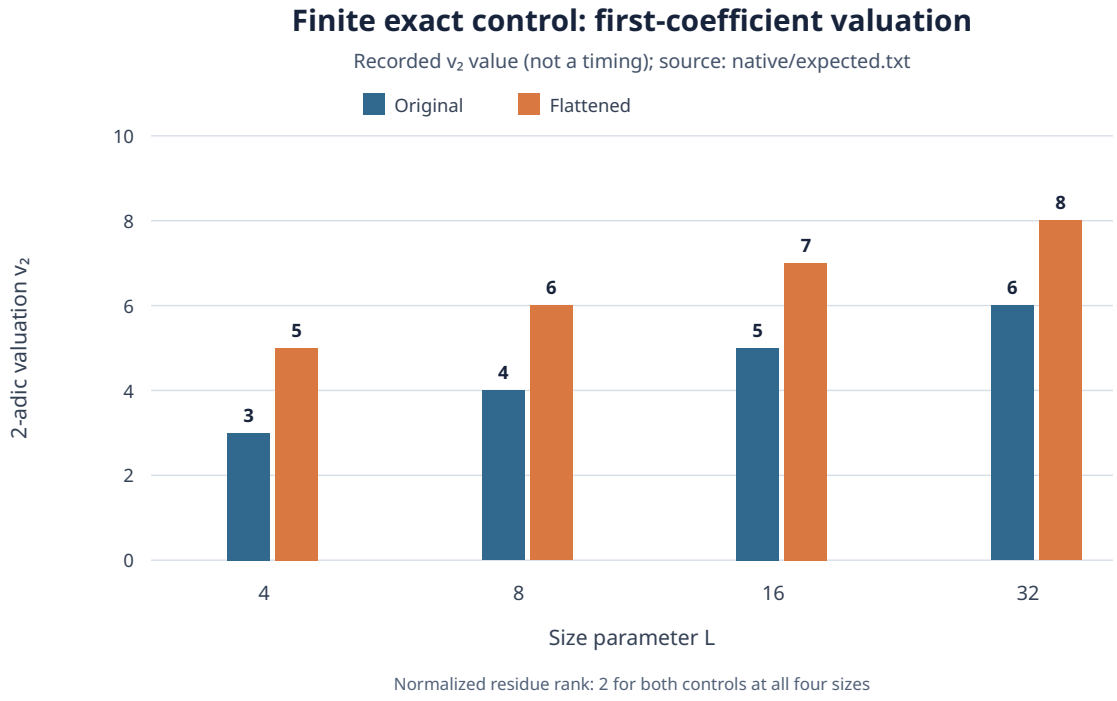


Figure 1. Exact recorded v_2 by L from the native control. Each point is one size, not a timing sample. Normalized residue rank is 2 for both series at every size. The source SVG is `valuation.svg`.

Proposals, separately

The study proposes (1) replacing the scalar origin channel with a nonscalar matrix $R(T)$ whose off-node values vanish and whose crossing jets have low weighted rank, and (2) allowing free jets through $W_{new}(T) = W_{base}(T) + C(T)H(T)$. Neither construction has a supplied integral factorization certificate or a measured cost here. The finite flattening control is not evidence that either proposal works.

The next discriminating step is a bounded search that returns the candidate polynomial and entry-by-entry integral U, V witnesses at every retained node and stipulated precision. Its candidate domain is $\deg H \leq 2L - 1$; initial sizes are $L = 4, 8$, precisions $8, 12, 16$, and controls are $H = 0$, the recorded flattening construction, and rejection of an invalid even-denominator inverse. Before searching, freeze rational coefficient and denominator budgets and price polynomial degree and precision. A successful small case would still need a uniform construction and a complete recursive proof.

The linked backlog card `XP-20261101-rank-directed-jet-completion` is draft, unfunded, and not scheduled. The runner, frozen protocol, resource estimate, and authorization to execute do not exist yet.

Sources and limits

- Study README: proposals, control interpretation, formal scope, and deferred search inputs.
- `native/PROTOCOL.txt`: four-size exact-rational control protocol; no timing run.
- `native/expected.txt` and `native/replay.log`: recorded values and replay.
- `lean/PROOF_MAP.md`: formalized own-node value and first-derivative identities and their limits.

No new experiment, Lean build, ECDLP instance, or solver execution was performed for this report.